

# Determination of Serum Cholesterol by Near-Infrared Reflectance Spectrometry

E. Peuchant, C. Salles, and R. Jensen\*

Laboratoire de Biochimie, Hôpital Saint-Andre, 1, Rue Jean-Burguet, 33075 Bordeaux Cedex, France

**A new approach using near-infrared reflectance spectrometry for measuring serum cholesterol is reported. The method requires a calibration performed on 30 human sera samples containing cholesterol concentrations ranging from 3 to 12 mmol/L. A multiple linear regression makes it possible to select a filter combination that is characteristic of the matrix and its composition. The calibration constants calculated by the regression calculations are used to quantify serum cholesterol according to a mathematical equation. The method is reproducible and the results correlate very well with those obtained with the reference method (Liebermann-Burchard) giving a correlation coefficient of 0.963 and of 0.985 with the enzymatic method (Trinder assay). The present method provides excellent accuracy and gives a direct result without any reagent.**

Analysis of total serum cholesterol is very important in pathology and particularly in cardiovascular diseases (1-3). Several methods have been used to determine serum cholesterol and these may be divided into two groups: colorimetric methods (4, 5) and enzymatic methods (6-8). The lack of specificity and sensitivity of colorimetric methods has been offset by the specificity obtained with enzymes as reagents. However, inaccurate values are frequently obtained (9, 10) and the high cost of these tests is well-known owing to the specificity of the enzymes.

Our method is a new concept based on the near-infrared reflectance widely used in the food and agricultural industries (11-17). This infrared reflectance method, applied to serum cholesterol determination, requires preliminary calibration performed on human sera whose cholesterol concentration is measured with a reference chemical analysis. The calibration makes it possible to select characteristic wavelengths where the parameter, cholesterol, has specific absorption bands in the near-infrared region. The measurement of reflectance of a sample is in relation with the parameter concentration through a mathematical equation.

Near-infrared reflectance analysis (NIRA) is directly performed on serum without the need for extraction or reagent. The result is obtained immediately in less than 1 min per sample. The method is simple, rapid, and inexpensive and is in good correlation with reference and enzymatic methods.

## EXPERIMENTAL SECTION

**Apparatus.** Experiments were conducted by using a Technicon InfraAlyzer Model 450 interference-Filter-based reflectance spectrometer (Technicon Corp., Tarrytown, NY) connected to a Hewlett-Packard HP 86B microcomputer.

This apparatus was fitted with a tungsten halogen lamp and 19 narrow bandwidth (10 nm) filters of fixed wavelength specification covering from 1445 to 2345 nm. High spectral sensitivity was claimed through the use of Kohler optics and an integrating sphere to increase the light grasp (18).

The optical system of InfraAlyzer 450 represented in Figure 1 was divided into three parts.

(a) The light of the lamp which went parallel through an optical condenser, then struck a chopper, in order to modulate the light and to give a squared signal of high frequency allowing an absolute measurement of reflectance levels to total absence of light, and finally passed through filters placed on a disk which gave a monochromatic beam when in rotation.

(b) A classical optic of Kohler led the incident radiation alternatively to the sample (full line) and to photoelectric cells (dotted line) through a tipping mirror (Figure 1).

(c) A gold-plated integrating sphere with two circular apertures of 26 mm diameter, covered with a quartz slide on the two poles, allowed the incident beam to directly strike the sample or the photoelectric cells according to the position of the mirror. The integrating sphere collected the total beam reflected by the sample, while the specular beam which was perpendicular to the axis of the apertures returned to the lamp without being collected or measured by the cells. The specular beam resulted from reflection on areas of the sample or on the bottom of the liquid cell when the sample used was transparent to infrared radiation.

The apparatus made it possible to measure reflectance ( $R$ ) as the ratio of reflected energy from the sample and the incident radiation. In fact, the measurement of  $R$  was converted into  $\log(1/R)$  and the latter was measured at 19 wavelengths and entered into the computer. The principle of the measurement depended on a linear mathematical algorithm associating sample concentration with the different reflectance values.

The algorithm was as follows:

$$C_{1\text{mmol/L}} = a_0 + a_1 \log \frac{1}{R_{1-1}} + \dots a_n \log \frac{1}{R_{n-1}}$$

The calculation of the constants  $a_0, \dots, a_n$  was performed with a training set having precise rating and covering the whole range of cholesterol concentrations.

These concentrations must be made with a precise reference technique since NIRA accuracy depends upon that of the reference technique (19).

This gives a matrix of the type

$$C_{1\text{mmol/L}} = a_0 + a_1 \log \frac{1}{R_{1-1}} + \dots a_n \log \frac{1}{R_{n-1}}$$

$$C_{2\text{mmol/L}} = a_0 + a_1 \log \frac{1}{R_{1-2}} + \dots a_n \log \frac{1}{R_{n-2}}$$

$$C_{p\text{mmol/L}} = a_0 + a_1 \log \frac{1}{R_{1-p}} + \dots a_n \log \frac{1}{R_{n-p}}$$

The method of linear regression (20) was used to determine the values of the constants. The principle of the method was the following: The sum of squares of the residual values (difference between reference values and calculated values) must be as low as possible. Generally the training set requires some 30 samples; thus, the matrix is very important and a microcomputer is necessary to calculate the constant values. The TECHNICON (Domont, France) program was used to select the best combination of filters from a group of filters that were highly correlated spectroscopically: Wavelengths with clearly assignable spectral responses were included and those without were excluded. Wavelengths with low constants and poor correlations were excluded on statistical grounds by using Student's  $t$  test.

The program functioned with a combination of one, two, three, ..., 19 wavelengths giving for each combination the values of constants corresponding to each filter, the values of the correlation

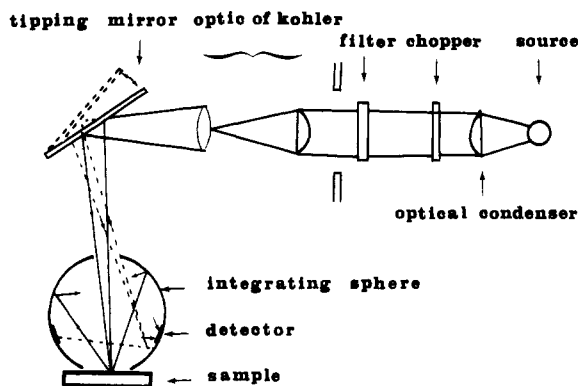


Figure 1. Optical system of an InfraAlyzer 450.

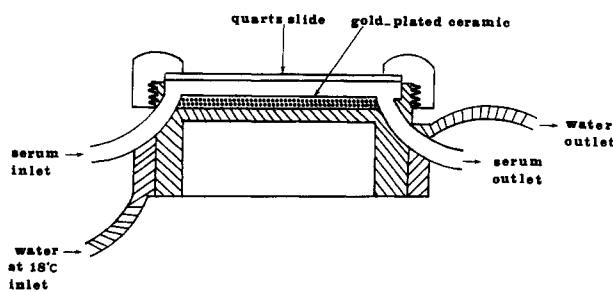


Figure 2. Thermostatically controlled liquid cell used for serum analysis.

coefficient  $R$ , and the constant of Fischer ( $F$  ratio) indicating the quality of the regression

$$R = \left( 1 - \frac{\text{SEE}^2(n-k-1)}{\text{SD}^2(n-1)} \right)^{1/2}$$

$$F \text{ ratio} = \frac{R^2(n-k-1)}{(1-R^2)k}$$

where SEE is the relative standard deviation, the standard error of estimate, SD is the standard deviation of the training set,  $n$  is the number of samples, and  $k$  is the number of filters.

When the number of wavelengths increased,  $R$  approached 1 and the  $F$  ratio increased. The regression calculations were stopped when  $R$  remained stable and the  $F$  ratio began to decrease.

**Training Set.** For the training set, 30 sera from hospitalized patients from St-André Hospital suffering from various diseases and following various treatments were selected to eliminate any possible interference due to endogenous substances (hemolysis, turbidity, ...) and common drugs. Only icteric sera were excluded since the Liebermann-Burchard method gives inaccurate values for these (21). To obtain better accuracy with the reference method, we performed duplicate cholesterol analysis on an automated SMAC (Technicon Corp., Tarrytown, NY).

Moreover, these sera covered the whole range of serum cholesterol from 3 to 12 mmol/L by incrementation of 0.3 mmol/L. Each sample was analyzed twice on the InfraAlyzer with another aliquot to avoid the risk of error due to nonhomogeneity of samples (19). When regression was achieved, the best wavelength combination with corresponding constants was entered into the apparatus.

The bias ( $a_0$ ) was used to adjust the calibration levels for differences in calculated values and standard values of cholesterol concentrations.

**Procedure.** The liquid cell was mounted on an interchangeable drawer connected to a cryostat, allowing temperature control at 18 °C prior to the entry of the sample into the cell. The liquid cell had a gold-plated ceramic bottom and a quartz window at the top limiting the volume to 100  $\mu\text{L}$  (Figure 2). Two chucks placed at each side of the cell made it possible to introduce the sample with a syringe and to eliminate the excess. Between two samples, the cell was cleared by air injection and an aliquot of the following sample was added to avoid contamination.

After selection on the apparatus of the analyzed product (serum) and the analyte (cholesterol), analysis was performed by

Table I. Results of Calibration with Five, Six, and Seven Selected Filters

| 5 selected filters<br>bias 27.666 |                    | 6 selected filters<br>bias 4.732 |                    | 7 selected filters<br>bias 48.709 |                    |
|-----------------------------------|--------------------|----------------------------------|--------------------|-----------------------------------|--------------------|
| filters                           | scaling<br>factors | filters                          | scaling<br>factors | filters                           | scaling<br>factors |
| 2                                 | 95.842             | 6                                | 567.630            | 7                                 | -926.830           |
| 5                                 | -142.56            | 7                                | -926.780           | 8                                 | 807.140            |
| 10                                | -71.614            | 8                                | 357.740            | 16                                | -8.120             |
| 18                                | 2638.100           | 17                               | -2922.300          | 17                                | -3271.400          |
| 20                                | -2596.400          | 18                               | 5162.500           | 18                                | 5873.600           |
|                                   |                    | 20                               | -2258.700          | 19                                | 108.770            |
|                                   |                    |                                  |                    | 20                                | -2744.300          |
| corr<br>coeff                     | 0.9205             |                                  | 0.9225             |                                   | 0.9257             |
| $F$ ratio                         | 127.2778           |                                  | 107.159            |                                   | 94.3321            |

Table II. Spectra Structure Correlations and Average Molar Absorptivity: Data Corresponding to Selected Filters

| filters | corresponding<br>wavelength,<br>nm | specific<br>corresponding<br>structure | av molar<br>absorptivity |
|---------|------------------------------------|----------------------------------------|--------------------------|
| 7       | 2208                               | <chem>CH=CH2</chem>                    | 1.2                      |
| 8       | 2190                               | -CH (aldehydic)                        | 0.5                      |
| 16      | 1940                               | -CH (aldehydic)                        | 0.5                      |
| 17      | 1734                               | -OH (water)                            | 1.2                      |
| 18      | 1722                               | -CH <sub>3</sub> CH <sub>2</sub>       | 0.1                      |
| 19      | 1445                               | -CH <sub>3</sub> CH <sub>2</sub>       | 0.1                      |
| 20      | 1680                               | -OH (water)                            | 0.7                      |
|         |                                    | -CH (aromatic)                         | 0.1                      |

pushing the drawer, and the value of cholesterol concentration appeared on the screen in less than 1 min.

## RESULTS AND DISCUSSION

Three sets of five, six, and seven filters were selected according to statistical results of multilinear regression (Table I). The combination of five filters corresponding respectively to 1680, 1722, 2180, 2270, and 2336 nm wavelengths presents a coefficient correlation of 0.9205. The set of six filters of 1680, 1722, 1734, 2190, 2208, and 2230 nm wavelengths, respectively has a coefficient correlation of 0.9225, and the set with seven filters (1680, 1445, 1722, 1734, 1940, 2190, and 2208 nm) has a coefficient of 0.9257. Thus the three sets of filters showed the same significant results: an identical correlation coefficient and an  $F$  ratio near 100.

These three sets of selected wavelengths were checked by analyzing unknown samples not used for calibration. The regression giving the best results of cholesterol concentration in comparison to the corresponding reference chemical values was selected. In our case, the wavelength set with the seven selected filters was selected. This regression was characterized by the highest correlation coefficient ( $R = 0.9257$ ) and an  $F$  ratio slightly lower than the others ( $F = 94.33$ ) but high enough to give a good regression. The calculated bias of the regression was adjusted, thereby allowing an increase of the correlation coefficient to 0.9456. The study of the selected filters in the three sets of selected wavelengths showed the common presence of two filters: filter 18 and 1722 nm and filter 20 at 1680 nm. Filter 18 appears to be very important because the corresponding scaling factor is the highest in the three regressions (Table I). This filter, as well as filter 17 (1734 nm), is correlated with molar absorptivity of specific groups such as methyl and ethyl groups in the near-infrared region (22) (Table II) which are present in serum cholesterol. At 1680 nm, groups such as -CH aromatic groups have a maximum absorption. The presence of a double bond in the structure

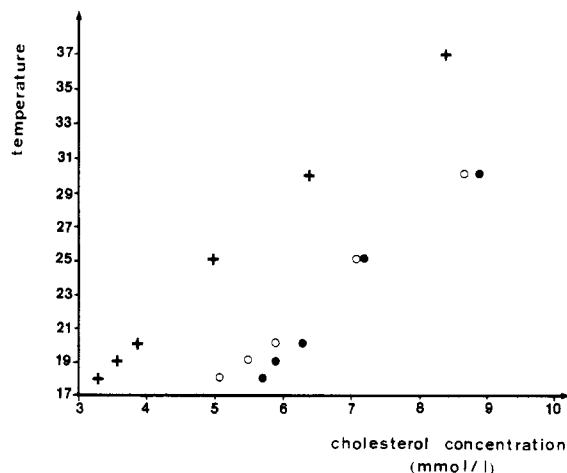


Figure 3. Influence of temperature on cholesterol concentrations.

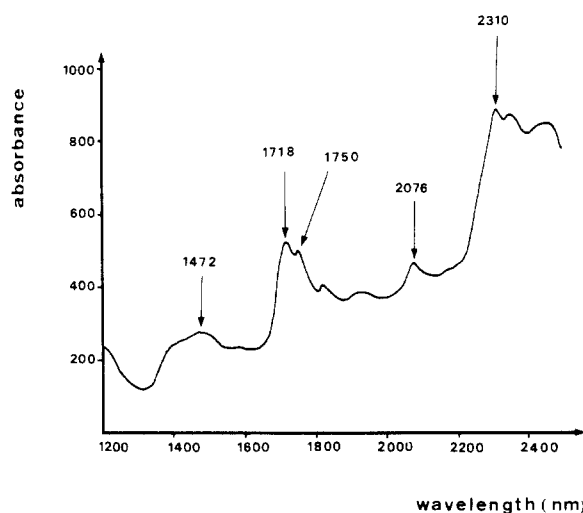


Figure 4. Representative spectrum of crystallized cholesterol in the near-infrared region (1200–2400 nm) performed on an InfraAlyzer 500 using micro absorbance units.

of cholesterol between  $C_5$  and  $C_6$  in cycle B probably involves the selection of filter 20.

The spectrum of crystallized cholesterol (Figure 3) analyzed on an InfraAlyzer 500, performing incremental steps of 2-nm scanning from 1200 to 2500 nm, confirms the important role of filters 17 (1734 nm), 18 (1722 nm), and 20 (1680 nm) with the presence of two peaks at 1718 and 1750 nm. As regards filters 16 (1940 nm) and 19 (1445 nm), these are related to absorbance of water hydroxyl groups (Table II). The spectrum of a human serum performed on an InfraAlyzer 500 and represented in Figure 5 shows the presence of two peaks at the wavelengths which correspond to the absorption of water.

Filters 7 (2208 nm) and 8 (2190 nm) are not related to any group present in the structure of cholesterol and seem to be selected in the mathematical equation to compensate the effect of possible interfering molecules. Therefore, the association of seven selected filters for measuring serum cholesterol makes it possible to eliminate all spectral interferences and avoids preliminary cholesterol extraction which is necessary in the near-infrared absorption spectrophotometry (23).

The study of the rise of temperature on the same sample significantly increases the calculated cholesterol concentration. Moreover, the study performed on three samples of different cholesterol concentrations indicates that the increase is not constant but depends on the cholesterol concentration at 18 °C (Figure 5). It is well-known that in the near-infrared region, a rise in temperature induces a modification in the fundamental vibrational bands.

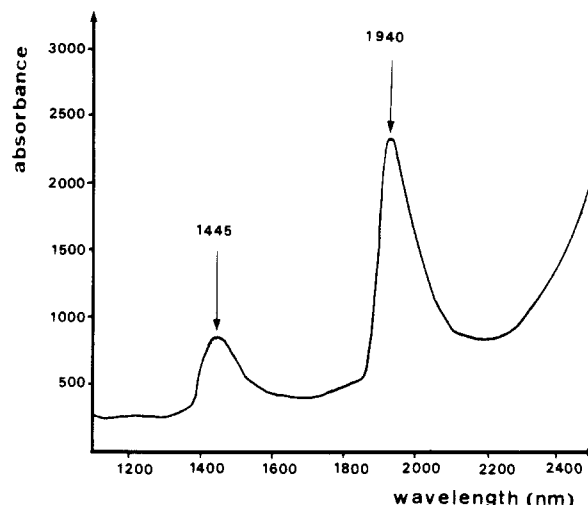


Figure 5. Representative spectrum of a human serum in the near-infrared region (1200–2400 nm) performed on an InfraAlyzer 500 using micro absorbance units.

This involves a broadening of the fundamental vibrational bands and hence a decrease in absorbance values, as may be seen on filters 17 (1734 nm), 18 (1722 nm), and 20 (1680 nm) which are the most representative of the cholesterol structure. Thus, the cholesterol concentrations, calculated from the above equations, increase with temperature. This is due to the negative scaling factors of filters 17 and 20, which are greater than the positive scaling factor of filter 18 (Table I). However, we have also noticed that the number of hydrogen bonds between cholesterol and water is lower for a low cholesterol concentration and decreases when the temperature rises. Therefore, the absorbance values recorded on filter 19, specific to hydroxyl groups (water), increase more with temperature for samples containing 3.3 mmol/L of cholesterol than for samples containing 4.8 and 5.5 mmol/L. This involves a higher increase of the calculated cholesterol value for the sample containing a low cholesterol concentration, because the influence of water diminishes. Therefore, the determination of a liquid sample requires a good reproducibility of temperature, of the order of a few tenths of a degree Celsius, in order to obtain the desired accuracy and reproducibility for analysis.

The accuracy of the method was also checked by plotting the values obtained by using a commercial cholesterol solution Seronorm (Fumouze, Nyegaard, Oslo, Norway) at a range from 0.74 to 7.4 mmol/L. The regression equation of the curve was  $Y = 0.128 \pm 0.966X$ , which indicates a great linearity and sensitivity; thus, very low levels of less than 1% of cholesterol may be detected.

The precision of this method was verified with two groups of sera, one including a pool of patient sera (5.05 mmol/L cholesterol) and the other one a commercial control serum (test point 1) (Technicon) containing 3.30 mmol/L of cholesterol. The coefficients of variation were, respectively, 2.17% and 2.00%, which are very acceptable.

In comparison with other methods (Liebermann–Burchard technique and Trinder assay) the new method is as sensitive. For this purpose, 60 sera from hospitalized patients were tested both with the NIRA method and these methods. The correlation with the reference method was studied on all patient sera except icteric sera. No interferences were found and the results obtained show a correlation of 0.965. The same study with the Trinder assay on sera including icteric sera showed an improved correlation of 0.982.

The new method is as specific, precise, and sensitive as the commonly used tests in biochemistry. However, the NIRA method is not destructive and requires no reagent for analysis.

Moreover, the method is relatively rapid: less than 1 min from sample introduction to cholesterol concentration value on screen. Certain modifications, in particular the automation of serum handling on the InfraAnalyzer which is an apparatus for industrial use, could increase rapidity. The quality of the results obtained with serum cholesterol makes it possible to measure other lipid serum analytes such as total lipids (24), phospholipids, triglycerides, and other biochemical analytes on the same 100- $\mu$ L sample and with the same method.

Registry No. Cholesterol, 57-88-5.

### LITERATURE CITED

- (1) Miettinen, M.; Turpeinen, O.; Karvonen, L. J. *Lancet* **1972**, *2*, 835-888.
- (2) Ducimetiere, P.; Claude, J. R.; Richard, J. L. *Arterial Wall* **1977**, *7*, 71-76.
- (3) Gordon, T.; Kannel, W. B. J. *Am. Med. Assoc.* **1972**, *221*, 661-667.
- (4) Touks, D. B. *Clin. Biochem.* **1967**, *1*, 12-17.
- (5) Fasce, C. F.; Vanderlinde, R. E. *Clin. Chem. (Winston-Salem, N.C.)* **1972**, *18*, 901-910.
- (6) Flegg, H. M. *Ann. Clin. Biochem.* **1973**, *10*, 79-84.
- (7) Allain, C. C.; Poon, L. S.; Chan, C. S. G.; Richmond, W.; Fu, P. C. *Clin. Chem. (Winston-Salem, N.C.)* **1974**, *20*, 470-475.
- (8) Haecckel, R.; Perlick, M. J. *Clin. Chem. Clin. Biochem.* **1976**, *14*, 411-414.
- (9) Cooper, G. R.; Ulman, M. D.; Hazlehurst, J. *Clin. Chem. (Winston-Salem, N.C.)* **1979**, *25*, 1074-1079.
- (10) Bachoric, P. S.; Wood, D. S. In *Hyperlipidemia: Diagnosis and Therapy*; Rifkind, B. M., Levy, R. I., Eds.; Grune and Stratton: New York, 1977; pp 49-51.
- (11) Massic, D. R.; Norris, K. H. *Trans. Am. Soc. Agric. Eng.* **1965**, *8*(1), 598-600.
- (12) Hart, J. R.; Norris, K. H.; Golumbic, C. *Cereal Chem.* **1962**, *39*, 94-98.
- (13) Ben-Gera, I.; Norris, K. H. *Isr. J. Agric. Res.* **1968**, *18*(3), 117-124.
- (14) Hrushka, W. R.; Norris, K. H. *Appl. Spectrosc.* **1982**, *36*, 261-265.
- (15) Honigs, D. E.; Freelin, J. M.; Hieftje, G. H.; Hirschfeld, T. B. *Appl. Spectrosc.* **1983**, *37*, 491-497.
- (16) Baker, D.; Norris, K. H. *Appl. Spectrosc.* **1985**, *39*(4), 618-622.
- (17) Watson, C. A. *Anal. Chem.* **1977**, *49*, 835A-840A.
- (18) Wetzel, D. C. *Anal. Chem.* **1983**, *55*, 1165-1176.
- (19) Mark, H.; Workmann, J. *Anal. Chem.* **1986**, *58*, 1454-1459.
- (20) Draper, N. R.; Mith, A. *Applied Regression Analysis*; 2nd ed.; Wiley: New York, 1981.
- (21) Tel, R. M.; Berends, G. T. J. *Clin. Chem. Clin. Biochem.* **1980**, *18*, 595-601.
- (22) Goddu, R. F.; Delker, D. A. *Anal. Chem.* **1960**, *32*, 140.
- (23) Kisner, H. J.; Brown, C. W. *Anal. Lett.* **1985**, *13*, 377-397.
- (24) Jensen, R.; Lukan, I.; Peuchant, E. *Bull. Soc. Pharm. Bordeaux* **1986**, *125*, 223-232.

RECEIVED for review October 27, 1986. Accepted April 6, 1987.

## Organic and Elemental Ion Mapping Using Laser Mass Spectrometry

Zbigniew A. Wilk and David M. Hercules\*

Department of Chemistry, University of Pittsburgh, Pittsburgh, Pennsylvania 15260

**A commercial laser mass spectrometer (LAMMA 1000) has been automated and programmed to produce ion maps with high lateral resolution. The sample is scanned in an  $x$ - $y$  mode and time-of-flight mass spectrometric analysis is performed for each matrix element. A user-selected mass is searched for the maximum intensity which is density coded and displayed as a four-sided polygon. A matrix consisting of  $31 \times 31$  data points can be completed in 20 min. Ion maps were obtained for control samples consisting of organic dyes deposited on a nitrocellulose substrate with transmission electron microscopy grids as a mask. Results from maps of molecular cations have shown that the maximum lateral resolution obtainable was 2.5  $\mu$ m, which was limited by the diameter of the laser beam. Gold and aluminum ion maps were obtained from the analysis of a bonding pad in an integrated circuit demonstrating that elemental information can also be obtained. Additionally, ion maps were obtained for an inclusion in a coal maceral. The maps showed that the inclusion was composed primarily of iron and sulfur. Organic material was detected in localized areas on the periphery of the inclusion.**

Laser mass spectrometry (LMS) is a valuable analytical tool currently used in many areas of research. The use of a laser as an ionization source has several important advantages over more conventional ionization techniques. The laser has the ability to ionize a wide variety of samples, including involatile and thermally unstable organics. Analysis usually can be performed without elaborate sample preparation; few charging

problems are encountered. Laser ionization produces both positive and negative ions such that unambiguous structure determination can often be accomplished. Another advantage associated with the laser is the ability to focus the beam to a small spot, on the order of micrometers. The focused beam can be positioned at any location on the sample, and a mass spectral analysis can be obtained from that area. The ability of the laser to be accurately positioned onto the sample with a small beam diameter allows its use as a microprobe.

The importance of microprobe mass spectrometric techniques lies in their ability to obtain chemical information from microvolumes through the analysis of molecular and/or structurally significant ions. Few methods currently available have the ability to analyze organic components in or on an organic matrix. Briggs et al. have used secondary ion mass spectrometry (SIMS) to map an organosilicon lubricant on an organic polymer (1). By use of an ion gun having a spatial resolution of 30  $\mu$ m, it was shown that SIMS can be used effectively as an organic microprobe provided that sample charging can be neutralized. Another mass spectrometric microprobe is the Cameca ion microscope (2, 3). The principal use of the Cameca has been in the area of elemental analysis. Organic analysis using this technique is difficult due to sample charging (coating methods are used to circumvent this) and the high primary ion flux which is necessary. Possible use of the Cameca as an organic microprobe has been demonstrated by using an ion beam (4) and a laser (5) for sample ionization.

Recently we have demonstrated the ability of the laser mass spectrometer to map organic ions from an organic matrix (6). These first experiments were designed to demonstrate the feasibility of imaging an organic molecular ion on an organic